B. Array

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: greedy | https://codeforces.com/problemset/problem/2209/B

You are given an integer array a of length n.

For each index i, find **the maximum number** of indices j such that j gt i and $|a_i - k|$ gt $|a_j - k|$, over all possible integer values of k.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The first line of each test case contains an integer n ($1 \le n \le 5000$).

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($-10^9 \le a_i \le 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed 5000.

Output

For each test case, output n integers denoting the answer.

Example

Input
6 1 1092 2 105 -105 5 1 2 93 84 2 7 2 9 38 4 7 1 6 10 1 9 20 9 829 3 87 1 283 7 11 9 18 29817 283 3 3928 5726 1942 1000000000 -1000000000 19

Output
0 1 0 4 2 2 1 0 5 4 4 2 2 1 0 8 4 4 3 5 3 2 2 1 0 8 7 7 4 5 3 3 2 2 1 0

Note

In the second test, the answers are:

- For i=1, you can choose k=-195, then j=2.

- For i=2, you can choose k=5, there exists no index j gt i.

In the third test, the answers are:

- For i=1, you can choose k=195, then j=2,3,4,5.
- For i=2, you can choose k=78, then j=3,4.
- For i=3, you can choose k=15, then j=4,5.
- For i=4, you can choose k=15, then j=5.
- For i=5, you can choose k=998 244 353, there exists no index j gt i.